

POLS 3316: Statistics for Political Scientists

Homework 1: Chapters 1–3

Due Tuesday, September 15 at 11:59 PM · 100 points

Name: _____

Instructions

Multiple Choice: Questions 1–25 are worth 4 points each. Choose one answer for each question.

You may use the textbook, your notes, and the lecture slides while completing this homework.

Record your final answers on the Answer Sheet on the last page.

Submission

You may complete the homework in either of the following ways:

1. **Print the homework and complete it by hand.** Then scan or take clear photos of all completed pages.
2. **Complete the homework electronically** by writing or marking your answers directly on the PDF.

When you are finished, **upload your completed homework to Canvas by the deadline.**

Make sure your **name is included** and that all answers are clearly visible before submitting.

Chapter 1: The Scientific Study of Politics

1. [4 points] Which statement best explains why political scientists use theory instead of relying only on facts?

- A. Facts are always more accurate than theories.
- B. Theory helps explain why events happen and identify patterns that may apply beyond one case.
- C. Theory tells us exactly what will happen in every future election.
- D. Facts are not useful for studying politics.

2. [4 points] Which statement is a hypothesis?

- A. A stronger economy helps the incumbent party because voters reward good economic performance.
- B. Higher economic growth will be associated with a higher incumbent-party vote share.
- C. Economic performance is an important political concept.
- D. Incumbent parties should manage the economy well.

3. [4 points] Which variable is the outcome that we want to explain?

- A. Independent variable (X)
- B. Dependent variable (Y)
- C. Control variable (Z)
- D. Hypothesis

4. [4 points] We ask: "Does the number of hours studied affect exam scores?" Which answer correctly identifies X and Y?

- A. X = exam score; Y = hours studied
- B. X = hours studied; Y = exam score
- C. X = student motivation; Y = hours studied
- D. X = exam difficulty; Y = student motivation

Use Figure 1 for Question 5.

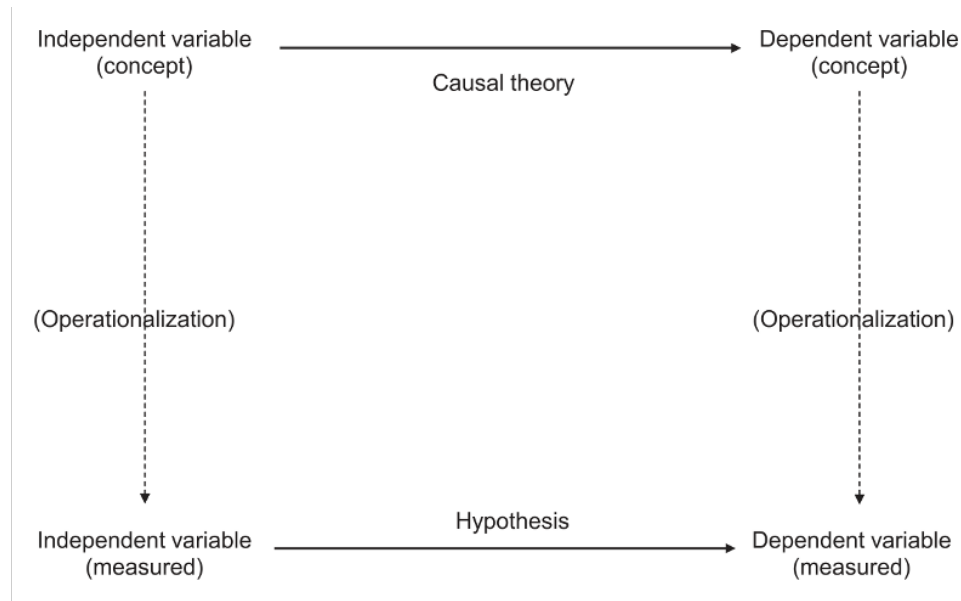


Figure 1. From concepts to measured variables (figure used in class).

5. [4 points] Look at Figure 1. What does operationalization mean?

- A. Deciding which variable causes another variable
- B. Turning an abstract concept into something we can actually measure
- C. Comparing two theories
- D. Predicting the exact outcome of an election

6. [4 points] We measure political participation by counting how many political activities a person did. A respondent did two activities and receives a score of 2. What does the score of 2 represent?

- A. The respondent's measured level of political participation
- B. A causal theory
- C. An independent variable
- D. A hypothesis

7. [4 points] In class, we used economic growth as one measure of economic performance. If growth goes up and incumbent-party vote share also tends to go up, the relationship is:

- A. Positive
- B. Negative
- C. Spurious
- D. Normative

8. [4 points] Which statement best describes the difference between correlation and causation?

- A. Correlation means X causes Y.
- B. Correlation means X and Y move together, while causation means X makes Y change.
- C. Correlation and causation mean the same thing.
- D. Causation means X and Y are measured at the same time.

Chapter 2: The Art of Theory Building

9. [4 points] Which question gives us a clear variation in Y to explain?

- A. I am interested in voter turnout.
- B. Voting is important in a democracy.
- C. Why do some people vote while others do not?
- D. Everyone should vote.

10. [4 points] Which question is written at a more general theoretical level?

- A. Why did the U.S. economy hurt Jimmy Carter in 1980?
- B. How do economic conditions affect national elections?
- C. Was inflation high before the 1980 election?
- D. Did Jimmy Carter campaign enough in 1980?

Use Figure 2 for Questions 11–12.

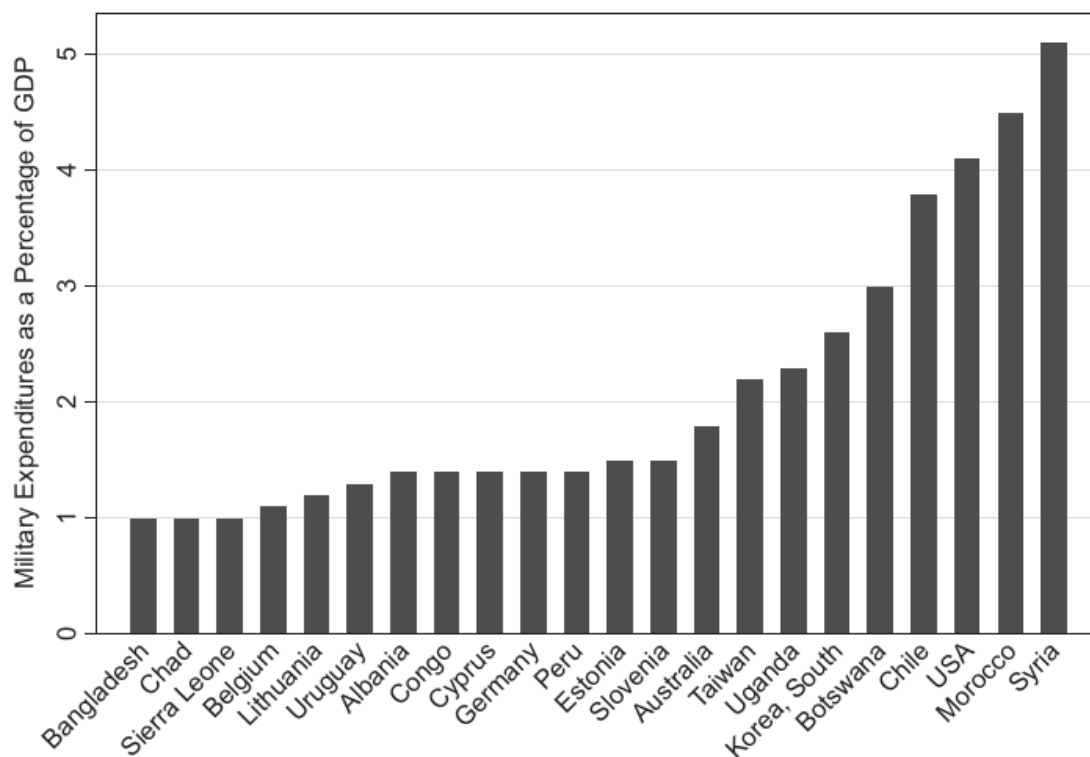


Figure 2.1 Military spending in 2005

Figure 2. Military spending across countries in 2005 (Kellstedt & Whitten, Chapter 2).

11. [4 points] Look at Figure 2. Military spending is measured for several countries in 2005. What type of measure is this?

- A. Time-series
- B. Cross-sectional
- C. Experimental
- D. Deterministic

12. [4 points] In Figure 2, what changes and what stays fixed?

- A. The years change; the country stays fixed.
- B. Both country and year change.
- C. The countries change; the year stays fixed at 2005.
- D. Neither country nor year changes.

13. [4 points] Which example is a time-series measure?

- A. Military spending across 22 countries in 2005
- B. Presidential approval in the United States measured every month for ten years
- C. Voter turnout across the 50 states in one election
- D. Women's representation across 20 countries in 2004

14. [4 points] Education may increase political participation, but the effect could be stronger for younger people than for older people. In this theory, age is:

- A. X
- B. Y
- C. Z, a conditioning variable
- D. The null hypothesis

15. [4 points] Suppose earlier research explains trust in government using the economy and corruption. You argue that social media use is another cause of trust. What are you adding?

- A. A new Y
- B. A new X
- C. A new time dimension
- D. A new null hypothesis

16. [4 points] In the class example, the utility of coming to class is basically:

- A. Benefits minus costs
- B. Costs minus benefits
- C. Benefits plus costs
- D. Benefits times probability

17. [4 points] What is a key assumption of formal theory?

- A. People try to make choices that best serve their interests.
- B. People always agree with each other.
- C. Political choices are completely random.
- D. People never respond to incentives.

Chapter 3: Evaluating Causal Relationships

18. [4 points] Most causal relationships in political science are probabilistic. What does that mean?

- A. X guarantees Y in every case.
- B. X changes the likelihood of Y, but there can be exceptions.
- C. X and Y cannot be measured.
- D. The relationship never changes across cases.

Use Figure 3 for Questions 19, 22, 23, and 25.

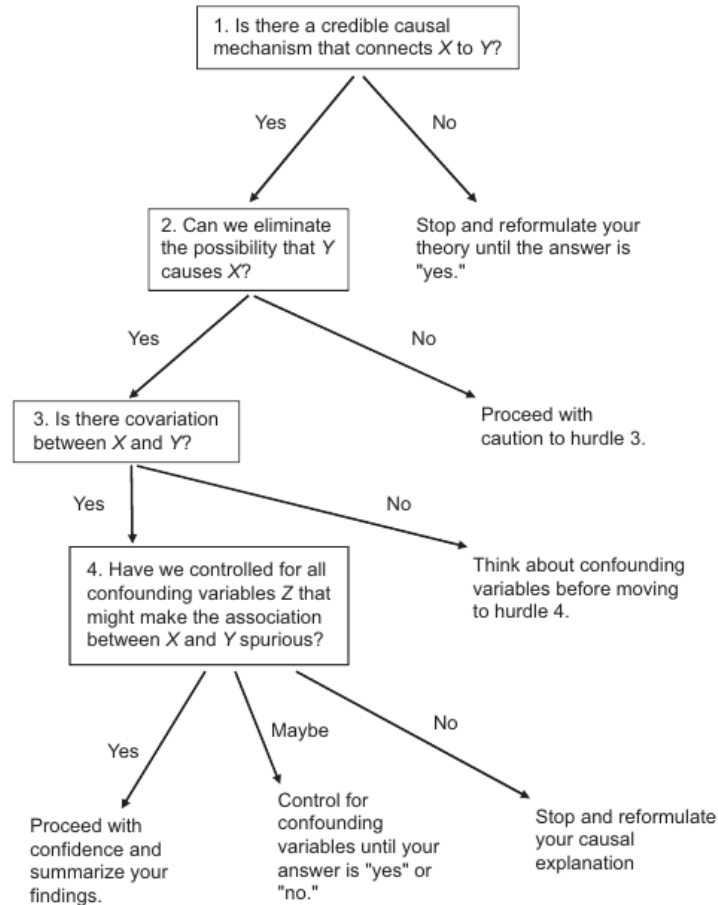


Figure 3.1 The path to evaluating a causal relationship

Figure 3. The four causal hurdles (Kellstedt & Whitten, Chapter 3).

19. [4 points] Look at Figure 3. What is the first causal hurdle?

- A. Is there a credible causal mechanism connecting X to Y?
- B. Do X and Y covary?
- C. Have we controlled for all confounding variables?
- D. Is the relationship positive?

20. [4 points] Which statement gives a credible causal mechanism for economic voting?

- A. Economic growth and vote share can both be measured.
- B. Voters evaluate the economy, hold the governing party responsible, and reward or punish it at the polls.
- C. Incumbent-party vote share is Y.
- D. Economic growth is X.

21. [4 points] Ice cream consumption and homicide both increase in hotter months. Why is “ice cream causes homicide” a weak causal claim?

- A. There is no believable causal mechanism connecting ice cream consumption to homicide.
- B. Homicide cannot be measured.
- C. Ice cream cannot be X.
- D. Any relationship over time must be causal.

22. [4 points] What is the second causal hurdle?

- A. Can we rule out the possibility that Y causes X?
- B. Is X measured as a percentage?
- C. Is the relationship positive?
- D. Does the theory contain only two variables?

23. [4 points] What is the third causal hurdle?

- A. Can we prove the theory is true?
- B. Do X and Y covary?
- C. Can we remove all measurement error?
- D. Is Y measured before X?

24. [4 points] Head Start participants tend to have better later educational outcomes. But highly involved parents may be more likely to enroll their children in Head Start and also help them do better in school. Here, parental involvement is:

- A. Y
- B. A null hypothesis
- C. A confounding variable Z
- D. A measure of Head Start participation

25. [4 points] What is the fourth causal hurdle?

- A. Have we controlled for confounding variables that could affect both X and Y?
- B. Is X larger than Y?
- C. Is the relationship negative?
- D. Can we make the theory more complicated?

Answer Sheet

Name: _____

Write A, B, C, or D for each question.

- | | | | | |
|-----------|-----------|-----------|-----------|-----------|
| 1. _____ | 2. _____ | 3. _____ | 4. _____ | 5. _____ |
| 6. _____ | 7. _____ | 8. _____ | 9. _____ | 10. _____ |
| 11. _____ | 12. _____ | 13. _____ | 14. _____ | 15. _____ |
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| 21. _____ | 22. _____ | 23. _____ | 24. _____ | 25. _____ |